

MAX HARPER



SUMMARY

Data analyst and quantitative researcher with experience across investment management, FMCG analytics, and financial data science. Proficient in Python with hands-on experience in machine learning, cloud computing, and AI-driven automation. Demonstrated financial experience developing and backtesting trading strategies, optimising portfolios, and supporting investment processes. Passionate about leveraging data and technology to solve challenging problems.

CONTACT DETAILS

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PERSONAL INFORMATION

Year of birth: 2002
Citizenship: Australian and Canadian

SKILLS

- Python · pandas · sklearn · PyTorch
- SQL · R · Excel · Git
- AWS & GCP (servers and serverless)
- Data visualisation & communication
- Quant strategy research & backtesting

HOBBIES

- Surfing, Rugby, Tech

EDUCATION

- **Sydney Grammar School:** Completed HSC in 2020 with 99.85 ATAR.
- **University of Sydney:** Completed **Bachelor of Advanced Computing and Commerce (Honours)**, majoring in Data Science and Business Analytics. Honours thesis: forecasting VIX futures using machine learning methods.
 - **First Class Honours**
 - **High Distinction 86 Honours WAM.**
 - **Engineering Entry Undergraduate Scholarship.**
 - **Daylell Scholarship.**
 - **Academic High Honour Roll.**
 - **Dean’s List for Excellence in Academic Performance.**
 - **Discipline of Business Analytics Prize in 3rd Year Quantitative Business Units.**

RELEVANT PROJECTS

- **VIX Futures Forecasting via Implied Volatility Surface (Honours Thesis):** Investigated whether the S&P 500 implied volatility surface contains predictive information for VIX futures. Applied PCA and autoencoders to compress the high-dimensional IV surface into forecasting features, evaluated across multiple ML models. Trading simulations on VIX futures demonstrated economically significant profits.
- **Investment Process Automation:** Developed a GCP-based system to automate core investment management processes, including risk parameter calibration, monitoring, and reporting. Leveraged LLMs to convert unstructured data into structured insights for investment governance.

EMPLOYMENT HISTORY

- **Quantium CBAiQ - Graduate Analyst** **2026.02-Present**
Graduate analyst in CBA’s analytics division, applying machine learning and data science to deliver insights at scale.
- **NJR Partners - Investment Technology Analyst** **2024.02-2026.01**
Automated investment process systems in the cloud as described above as well as implementation of AI tools to streamline investment process.
- **Bishopsgate Capital - Quantitative Analyst** **2025.01-2025.11**
Developed and backtested systematic trading strategies in volatility derivatives, leveraging the SPX implied volatility surface and machine learning models to forecast VIX futures. Research conducted in parallel with Honours thesis.
- **Blue Lake Partners - Quantitative Analyst** **2024.05-2024.12**
Researched and backtested systematic strategies in commodity equities, including signal generation and portfolio construction across the sector.
- **Fulton Market Group - Data Analyst Intern** **2023.12-2024.02**
Contributed to agricultural commodity price forecasting, as detailed in the project section.
- **Quantium - Data Analyst Intern** **2022.12-2023.02**
Worked in the fast-moving consumer goods division, conducting data investigations and backtesting an analytical model. Gained experience in collaborative teamwork within an agile business environment.